

MD SHARIFUL ALAM

Jamaica, NY 11435 | 516-808-3987 | ae.shariful@gmail.com | [linkedin.com/in/sharifulalam1](https://www.linkedin.com/in/sharifulalam1)

EDUCATION

New York Institute of Technology, Old Westbury, NY

Expected: December 2026

B.S. in Mechanical Engineering, Minor in Mathematics • CGPA: 3.99/4.00

Honors: Presidential Honor List (7/7 semesters), CoECS Dean's Recognition Scholarship (2024, 2026), T.K. Steele Scholarship

Relevant Coursework: Thermodynamics, Heat Transfer, Machine Design, Vibrations and Systems, Fluid Mechanics, Propulsion, Intro to CAD, Material Science, Computational Methods, Manufacturing, Mechatronics

Professional Organizations: American Society of Mechanical Engineers (ASME), National Society of Black Engineers (NSBE)

TECHNICAL SKILLS

- **Engineering Software:** MATLAB, C#, Multisim
- **CAD Software:** SolidWorks, Inventor, AutoCAD
- **FEA Software:** Abaqus, HyperMesh, Blender
- **Other Software:** MS Word, MS Excel, MS PowerPoint, Adobe Photoshop, Adobe Illustrator

RESEARCH EXPERIENCE

Research Assistant — Dr. Wei Zeng, NYIT

May 2026 – Present

- Conducting NSF-funded finite element analysis (FEA) of bone-screw implant pull-out mechanics in Abaqus and HyperMesh, investigating the effects of trabecular bone porosity, cortical shell presence, and screw geometry on implant fixation strength.
- Reviewing peer-reviewed publications on bone-implant biomechanics and FEA simulation techniques to identify research gaps, inform model assumptions, and support the study's computational framework.

Research Assistant — Dr. Tindaro Ioppolo, NYIT

September 2025 – April 2026

- Collaborated with a faculty researcher under NYIT's Undergraduate Research and Entrepreneurship Program (UREP) to redesign and enhance an optical microsphere-based wall shear stress sensor using the Whispering Gallery Mode measurement principle.
- Optimized sensor fabrication and testing procedures to improve measurement accuracy and sensitivity.

PROJECTS

Half-Car Suspension System Modeling and Simulation — NYIT

Fall 2025

- Developed a 4-DOF suspension model in MATLAB from first principles; tuned parameters across 4 runs to meet 2 of 3 ride-comfort specs, cutting pitch rate from 0.57 to 0.50 rad/s and settling time from 3.69 to 2.82 s.

Material Selection for Wind Turbine Blades — NYIT

Fall 2024

- Applied Ashby Selection Methodology across 5 screening criteria to narrow GRANTA EduPack's Level 2 database from 9 material categories to 4 finalists, selecting CFRP as the optimal blade material.

Paper Bridge Structural Design Competition — NYIT

Spring 2024

- Led a 3-person team to design and build a Howe truss bridge, achieving 206 kg load capacity and a force-to-weight ratio of 187 — top structural efficiency in the class.

EXPERIENCE

Peer Tutor — NYIT Office of Academic Success and Enrichment, Old Westbury, NY

March 2024 – Present

- Tutor ~15 undergraduates across 2–3 weekly sessions in manufacturing, thermodynamics, CAD, MATLAB, physics, and circuits; track individual progress and adapt session plans based on recurring error patterns.

Undergraduate Math Tutor — NYIT Math Resource Center, Old Westbury, NY

October 2025 – Present

- Tutor 6 students across 2 weekly sessions in Precalculus through Differential Equations and Linear Algebra; log attendance and progress after every session.

Accounting Assistant — Hasan Tax & Accounting Inc.

March 2024 – Present

- Assist in preparing individual and business tax returns, process payroll forms, and support bookkeeping tasks.
- Maintain and organize financial and client records; assist with immigration documentation.